CpE 187 Final Project Report

Stability and Level Monitoring Device

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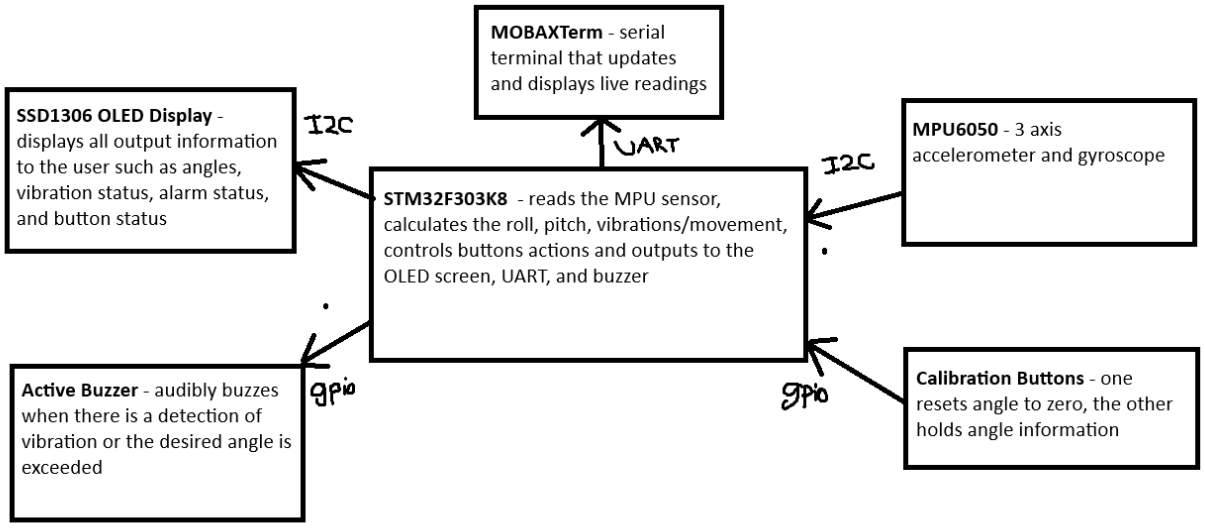
Team members and responsibilities

Yousef Ahmadzai – designed the project, programmed, troubleshooted, and built the device

Project Description

For this project, I designed a stability and level monitoring device. The device uses an STM32, MPU6050, OLED display, and an active buzzer to accurately read, output, and warn differences in angle and vibration/stability when measuring level. The MPU6050 is responsible for measuring any axis changes or vibrations, which is then used to calculate the angle that the device is at. The OLED displays show measurements to the user in real time and the MPU and OLED communicate together through a shared I2C bus with the STM23. When an angle/movement is exceeded, the buzzer plays an audible warning sound. There are also two buttons to use for calibration purposes of resetting the level to zero at the desired angle if needed, as well as a hold feature that lets the angles currently being displayed become locked so that the user may keep note of them until the button is pressed again and the real time measurement is resumed. This is all done through UART transmission.

Block Diagram

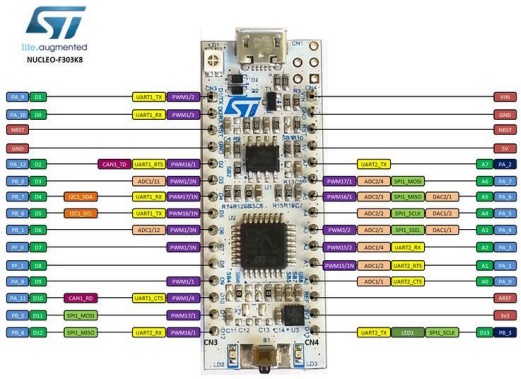


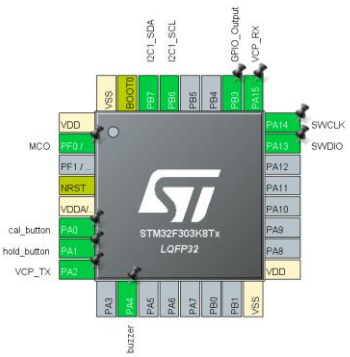
Background

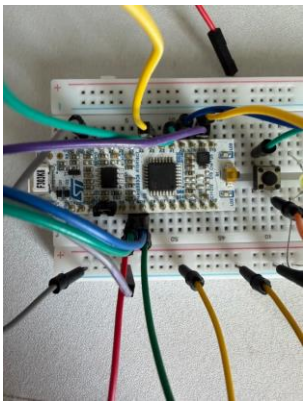
This section should go over sensors or electrical mechanical devices (i.e. motors).

[STM32F303K8]

This is the main microcontroller, without this the device would not be able to be created. It reads all the sensors and devices, does the calculations and measurements, allows each device to communicate with each other, and outputs to where it is programmed to. It uses a shared I2C bus for this project between the MPU and OLED display, as well as GPIO and UART in order to display information and take in decision made by the user such as button presses. The pins used for this are shown in the screenshot below. The pins match as such: PB7 is the SDA data line shared by the MPU and OLED, PB6 is the SCL clock line shared between the MPU and OLED, PA15 and PA2 are UART connection pins needed to output information, PA0 is the calibration/reset level button, PA1 is the hold angle button, PA4 is the active buzzer pin, the rest are defaulted from the current settings in the STM32.







[MPU6050]

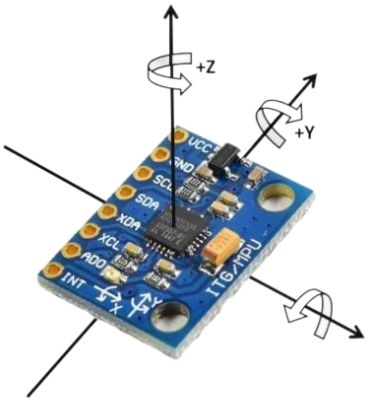
This is a three axis accelerometer and three axis gyroscope so it basically can read movement forwards, backwards, up, down, and roll or pitch changes. The program reads these measurements through I2C and then converts them to angle measurements. The SCL and SDA are the same connections for the OLED too, and its powered by the VCC which goes to the STM32's 3v3 pin. The pitch and roll are calculated in the program through the effects of gravity on the different axes and the formulas used for this are the following:

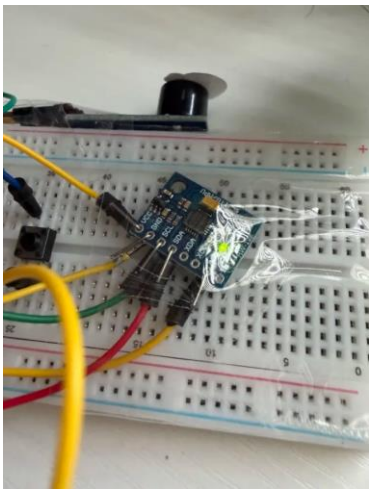
$$\text{Pitch} = \tan^{-1}\left(\frac{-a_x}{\sqrt{a_y^2 + a_z^2}}\right) \times \frac{180}{\pi}$$

rawPitch = atan2f(-(float)accelX, sqrtf((float)accelY * accelY + (float)accelZ * accelZ)) * 57.2958f;

$$\text{Roll} = \tan^{-1}\left(\frac{a_y}{a_z}\right) \times \frac{180}{\pi}$$

rawRoll = atan2f((float)accelY, (float)accelZ) * 57.2958f;

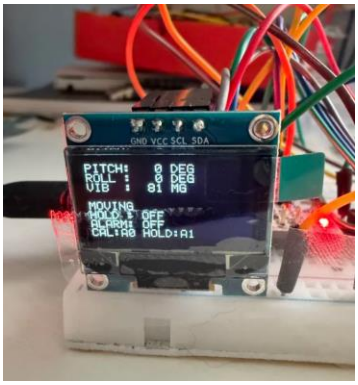




[SSD1306 OLED Display]

This is the screen used to display the information for this device. It has a resolution of 128x64 and like mentioned before this and the MPU have the same shared wiring to the STM pins. They have different addresses and that's what lets them basically use the same I2C bus though. It takes the serial terminal readings and outputs them to the user, specifically it shows the pitch and roll in angles, the vibration amount in milligravity, whether its stable, moving, or high vibration, and if the hold/calibrate buttons are pressed on or off. It also shows the pins that its connected to for those buttons at the bottom so they can be differentiated.

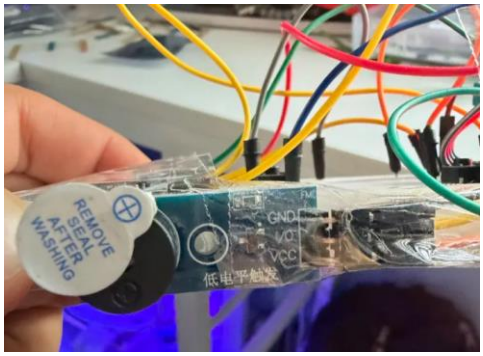




[Active Buzzer]

The buzzer produces a loud warning sound when there is detection of excess angle/tilt and vibration or movement. The reason its called an active buzzer is because it has an oscillator inside it that makes its own sound when the STM32 gives it the GPIO output. It could be programmed to different settings for when to buzz.





Results

Test Cases

Case 1 – Measuring of surface angle/level

For this test case it was a test to see if the device would accurately measure the angle of the surface it is placed on, and whether it would detect a change in angle and vibration or movement and whether it would update it in real time and alarm when the limit is exceeded. For this test I tried it with a limit of 4 degrees and then a follow up of 1 degree to show it can be precise. So when the pitch or roll exceeds the set value, it should beep and that is what it did. The video is below to open it double click it please.

trim.C2ABDA26-5ABD-46E2-B2A5-D8899BC8D2BE.mp4 first video is the 4 degree test

trim.7C1243DA-56D6-4D44-B6E2-A9CB14FC109F.mp4 second video is 1 degree test

Case 2 – Vibration/Stability detection

For this test case the device detects unnecessary movement and detects the condition that is currently active in the reading such as stable, moving, or high vibration. The sensitivity it was set to was to detect beyond 0.20g of movement or higher, one thing to note however, the buzzer responds instantly as the OLED takes a very small delay to update what it says. Here is the video.

trim.8422DDD1-2576-4008-AAC4-82AA56D8AE72.mp4

Case 3 – Active Calibration

In this test case it demonstrates how this device can be actively calibrated when needed meaning the zero can be reset to whatever level is needed and it will update the OLED in real time and the buzzer will associate the detection with that new zero, as well as being able to hold the information on the screen output if that button is pressed. The OLED also shows the buttons status of off and on too when this is happening.



trim.6F97FD6C-EF16-447E-AE7F-FB4A460A905D.mp4

Conclusion

Overall with the finished state of the project, I was satisfied with how it turned out. I didn't realize how precise it could get down to measuring which I liked how it ended up becoming. All three of my test cases passed and I learned that there is actually lots you can do with such simple small devices when they work together. Something that was definitely different than originally planed was my idea to have a LED indication system to show green, yellow, red depending on how close you are to the limit angle but that seemed to be not needed since the buzzer gives sound and the OLED already tells you your exact angle and such information. Something that I could have improved on would be making it powered without being connected to the cable so that I can use it on a shelf or something that I would be installing. It could be done with a battery as well as cleaner wiring and casing so it's not messy.

References

https://www.st.com/en/microcontrollers-microprocessors/stm32f303k8.html

https://www.hestore.hu/prod_getfile.php?id=8301&srsltid=AfmBOorcR0zXiZN5_JjnFCn8LfHcVABi5OTiY3HmhINBZzpW4vZgVBzi

https://e2e.ti.com/cfs-file/__key/communityserver-discussions-components-files/791/SSD1306-Datasheet-for-096-OLED-_2800_1_2900_.pdf